

Appendix R7A: Dimensionality - Regularization

Penalized Models: Ridge and LASSO Regressions

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This Quarto document contains R code examples about predictive modeling methods based on regularization, also known as shrinkage or penalized models. I will describe three methods – Ridge Regression and LASSO, and Elastic Nets, which is a blend of Ridge and LASSO. These models aim to correct issues of dimensionality, such as severe multicollinearity. An important thing to note is that these methods shrink the regression coefficients intentionally to bias the predictor coefficients to improve the model variance. Naturally, the more we shrink the coefficients, the more we bias the model. Cross-validation will help us identify the optimal amount of shrinkage in when the model variance is minimized. It is important to note that the more we shrink the model coefficients, the less useful the model becomes for interpretation and explainability. For the corresponding conceptual material associated with these appendices, consult the main book [Predictive Analytics and Machine Learning for Managers](#).

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Technical Note: All procedures involving machine learning and cross-validation are based on random sampling. Therefore, results may change when you run the analysis multiple times, depending on the kind of random number generator `RNGkind()` and the seed selected. The results in this script may not match the outputs in the main text of the book. Follow the chapter and this script appendix independently. Let's first set the random number generator default

```
RNGkind(sample.kind = "default") # To use the R default RNG
```

Dimensionality

The issue of dimensionality is often referred to as **curse of dimensionality**. As you add more predictors and complexity to a model specification, it will generally fit the data better and most fit statistics will improve. In some cases, the fit may even be perfect. For example, if you grow a tree until it has the same number of branches as you have data points, the fit will be perfect because the model will touch every single data point. Similarly, if you fit a smoothing spline model, you can set your parameters so that the resulting regression model with a jagged line that touches every single point in the data set. However, such models are **over-fitting** and tend to not perform well with new data due to extreme variance, making fit statistics like R-squared, MSE and 2LL to improve **artificially**, giving the false impression that the model is good. However, if we CV test the model with different data, over-fitted models will have higher variance, and these predictions will not be as accurate.

Over-fitting and multicollinearity are two of the many problems of dimensionality. The methods presented in this section are ideally suited to address dimensionality issues. There are many types of methods that address and correct for dimensionality. The 2 types of methods we will cover in this class are: (1) Regularization or Shrinkage (covered in this script); and (2) Dimension Reduction (covered in the Appendix R7B).

Regularization (i.e., Shrinkage or Penalized) Methods.

Regularized, shrinkage and penalized models are the same thing and they involve artificially shrinking regression coefficients to reduce the effect of multicollinearity on the model's variance. Generally speaking, Ridge and LASSO work best when the number of predictors is quite large (even thousands) and the data suffers from severe multicollinearity.

When coefficients are shrunk, the model's **bias** increases. On the upside, shrinkage reduces the model variance. The amount of shrinkage is controlled by a **shrinkage factor** lambda, which is a **tuning parameter**, meaning that you can shrink coefficients to various degrees and then select the shrinkage that yields the best cross-validation testing results (i.e., best predictive accuracy and optimal model variance). In this section I describe two popular regularization approaches: **Ridge** and **LASSO** (Least Absolute Shrinkage and Selection Operator) regressions. They both aim to shrink the coefficients, such that the coefficients of the less important variables become very small and, therefore, have little influence on predictions. The magnitude of the shrinkage can be controlled with the shrinkage factor "Lambda", such that when **Lambda = 0**, both Ridge and LASSO results are identical to OLS or GLM model, that is, a regression model with no shrinkage. In contrast, when **Lambda = infinite**, both methods also yield the same results as the **Null** model because all regression coefficients are squashed to 0. The difference between Ridge and LASSO is that as Lambda grows, Ridge coefficients become smaller, but not 0, except when Lambda is infinite. In contrast, LASSO shrinks coefficients to 0 along the way, dropping their respective predictors from the model as Lambda grows.

Because Lambda causes the coefficients to shrink, the larger the Lambda, the more **biased** the regression coefficients are compared to OLS or GLM coefficients. Generally, as Lambda grows, Ridge and LASSO improve the model's predictive accuracy up to a point, after which the predictive accuracy declines. An optimal model is one with a Lambda value that minimizes the model's CV test deviance. However, if the analytics goal is interpretability, it would OK to select a model with a smaller Lambda that has an acceptable CV test deviance.

How do you know if your model is biased? One way to quickly inspect for bias is to put the regression coefficients for various shrinkage factors in columns, from no shrinkage to extremely large shrinkage, and observe how the coefficients change from column to column. If a coefficient is relatively constant and stable, then it has little bias. If a coefficient fluctuates widely and even changes signs, then it has a lot of bias. Overall, the closer the coefficients are in value to the OLS coefficients, the less biased the model is and, consequently the more interpretable.

IMPORTANT: with shrinkage methods like Ridge regression, p-values no longer relevant, because all coefficients are retained in the model, although shrunk (i.e., biased). The actual magnitude of the coefficients tells you how important the corresponding variable is in the prediction. Similarly, the R-squared values are not reported, but for quantitative outcome models, you can use the deviance ratio illustrated above as an R squared.

Technical Note: One way to get coefficients in Ridge or LASSO regression representing variable importance is to standardize all variables. Ridge and LASSO do this automatically, but then standardization the coefficients for the output. But if you standardize the variables yourself, you will get standardized coefficients, which are unaffected

by variable scales. Consequently, predictors with larger standardized coefficients will have more weight on the predictions, thus providing a measure of variable importance.

Ridge Regression

Let's run a Ridge regression with several Lambdas and find the best Lambda that minimizes the 10FCV MSE. We will be using the `glmnet()` function from the `{glmnet}` R package. Let's start by loading the necessary packages.

```
library(glmnet)
library(ISLR) # Contains the Hitters baseball player salary data set
# This data set has several missing values, let's omit them
Hitters <- na.omit(Hitters)
options(scipen = 4) # Minimize use of scientific notation
```

Technical Note: the `glmnet()` function has a different syntax than other functions like `lm()` and `glm()`. It requires defining an **X matrix** (of the predictors) and a **Y vector** (of the response values). That is, rather than using the familiar `y ~ x1 + x2 etc.` formula syntax, we will use the `(X, Y, etc)` syntax, where X is the model matrix of predictors and Y is the vector of outcomes. We will use the `model.matrix()` function in the `{stats}` package to create the model's predictor matrix. The `model.matrix()` function will also convert all categorical variables to their respective dummy variables.

**** Technical Note:**** Notice below that I added `[-1]` at the end of the `model.matrix()` function. This is necessary because the `model.matrix()` function will include a column full of 1's as the first column, which represents the intercept. But Ridge also renders an intercept. So, if you don't remove this first column of 1's, the Ridge model will have 2 intercepts and the coefficients will be slightly off.

Let's start by creating the **x** predictor matrix and the **y** outcome vector:

```
x <- model.matrix(Salary ~ ., data = Hitters)[-1]
x[1:10, 1:10] # Take a look at the first 10 rows and columns
```

	AtBat	Hits	HmRun	Runs	RBI	Walks	Years	CAtBat	CHits	CHmRun
-Alan Ashby	315	81	7	24	38	39	14	3449	835	69
-Alvin Davis	479	130	18	66	72	76	3	1624	457	63
-Andre Dawson	496	141	20	65	78	37	11	5628	1575	225
-Andres Galarraga	321	87	10	39	42	30	2	396	101	12
-Alfredo Griffin	594	169	4	74	51	35	11	4408	1133	19
-Al Newman	185	37	1	23	8	21	2	214	42	1
-Argenis Salazar	298	73	0	24	24	7	3	509	108	0
-Andres Thomas	323	81	6	26	32	8	2	341	86	6
-Andre Thornton	401	92	17	49	66	65	13	5206	1332	253
-Alan Trammell	574	159	21	107	75	59	10	4631	1300	90

```
y <- Hitters$Salary # y vector with just the outcome variable
y[1:10] # Take a look at the first 10 values
```

```
[1] 475.000 480.000 500.000 91.500 750.000 70.000 100.000 75.000
[9] 1100.000 517.143
```

The `glmnet()` function is used to fit, Ridge, LASSO and Elastic Net models. It uses the parameter `alpha =` to indicate which model to fit, with a value of **0** for a **Ridge Regression**, **1** for **LASSO** and any value between **0-1**

for Elastic Net. `alpha` is the proportional weight given to **LASSO** and `(1 - alpha)` is the proportional weight given to **Ridge**.

Technical Note: By default, `glmnet()` estimates models with 100 different Lambda values, ranging from very large to very small (in descending order). You can estimate less or more Lambda values with the `nlambda =` parameter)

```
options(scipen = 999) # Disable scientific notation
ridge.mod <- glmnet(x, y,
                   alpha = 0)
```

Note About GLM Models: To fit a Ridge regression with a binomial outcome (e.g., **Logistic**), use the parameter `family = "binomial"`. For **count** outcomes, use the parameter `family = "poisson"`. The interpretation of the resulting coefficients is the same as with Logistic and Count data models.

Let's take a quick look at the first 10 Lambdas and their **Deviance Ratios** (i.e., proportion of Null deviance explained, like an R Squared), rounded to decimals). The Lambdas are sorted from largest (close to the Null model) to smallest (close to OLS). Naturally, the explained variance of the model (relative to the Null model) increases as lambda decreases because we are getting closer to the OLS model. But this will change once we CV test the model.

```
round(cbind("Lambda" = ridge.mod$lambda,
           "Deviance Ratio" = ridge.mod$dev.ratio)[1:10, ],
      digits = 4)
```

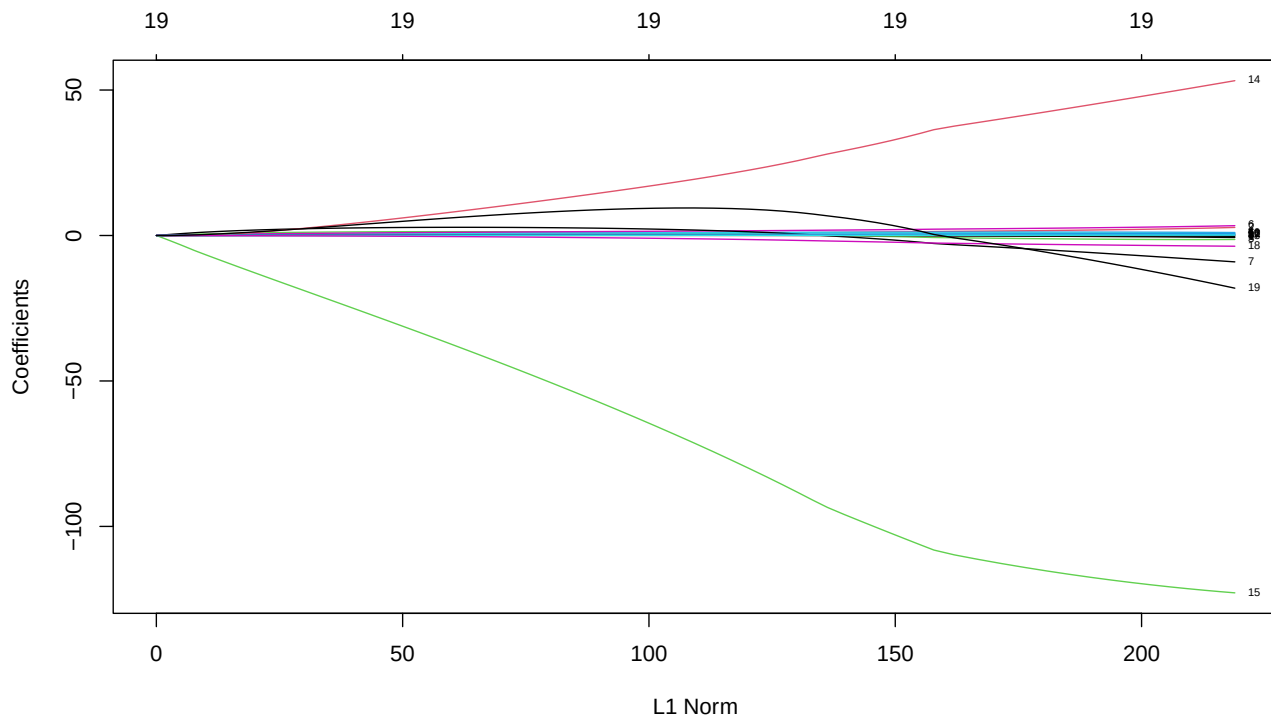
	Lambda	Deviance Ratio
[1,]	255282.1	0.0000
[2,]	232603.5	0.0116
[3,]	211939.7	0.0127
[4,]	193111.5	0.0139
[5,]	175956.0	0.0153
[6,]	160324.6	0.0167
[7,]	146081.8	0.0183
[8,]	133104.3	0.0200
[9,]	121279.7	0.0219
[10,]	110505.5	0.0239

Note: Ridge regression standardizes the coefficients because the scale of the variables can affect the proportion of shrinkage across coefficients. However, Ridge regression can be run without standardizing coefficients simply by specifying the parameter `standardize = FALSE`. The default is `TRUE`.

L2 Norm

L2 Norm is a measure of the overall shrinkage of the model. It is the square root of the sum of the squared coefficients in a Ridge regression. As Lambda increases, all coefficients shrink, so the L2 Norm decreases too. **L2 Norm = square root of the sum of the model's squared coefficients.** L2 Norm is mostly used as a preliminary visual analysis of the shrinkage pattern of the model. For example, notice in the graph below how the coefficients shrink. Some get pretty small fairly quickly, but no coefficient becomes 0 until Lambda is infinitely large. Also notice how some coefficients change in size more than others and some even change signs, providing some evidence of bias. Also, notice the repeated value of 19 at the top of the graph. This means that the model has 19 predictors at every shrinkage level. As we will see shortly, this will not be the case with LASSO.

```
# Note: the x axis is Labeled L1 Norm, but the prefix L1 is generally used for LASSO.
plot(ridge.mod, label = T)
```



To compute the **L2 Norm** for any lambda, for example 50 (notice we remove the first coefficient, which is the intercept):

```
l2.norm.50 <- sqrt(sum(coef(ridge.mod)[-1, 50] ^ 2))
l2.norm.50
```

```
[1] 22.53415
```

Optimal Shrinkage with CV Testing

Let's find the best Lambda shrinkage value that minimizes the CV test MSE using the `cv.glmnet()` in the `{glmnet}` R package. The syntax in the `cv.glmnet()` function is the same as for the `glmnet()` function, except that it has an additional parameters, `nfolds` = to change the CV test default from **10FCV** to other number of folds.

```
set.seed(1) # Arbitrary, to get repeatable results
cv.10Fold <- cv.glmnet(x, y,
                      alpha = 0) # 10FCV is the default

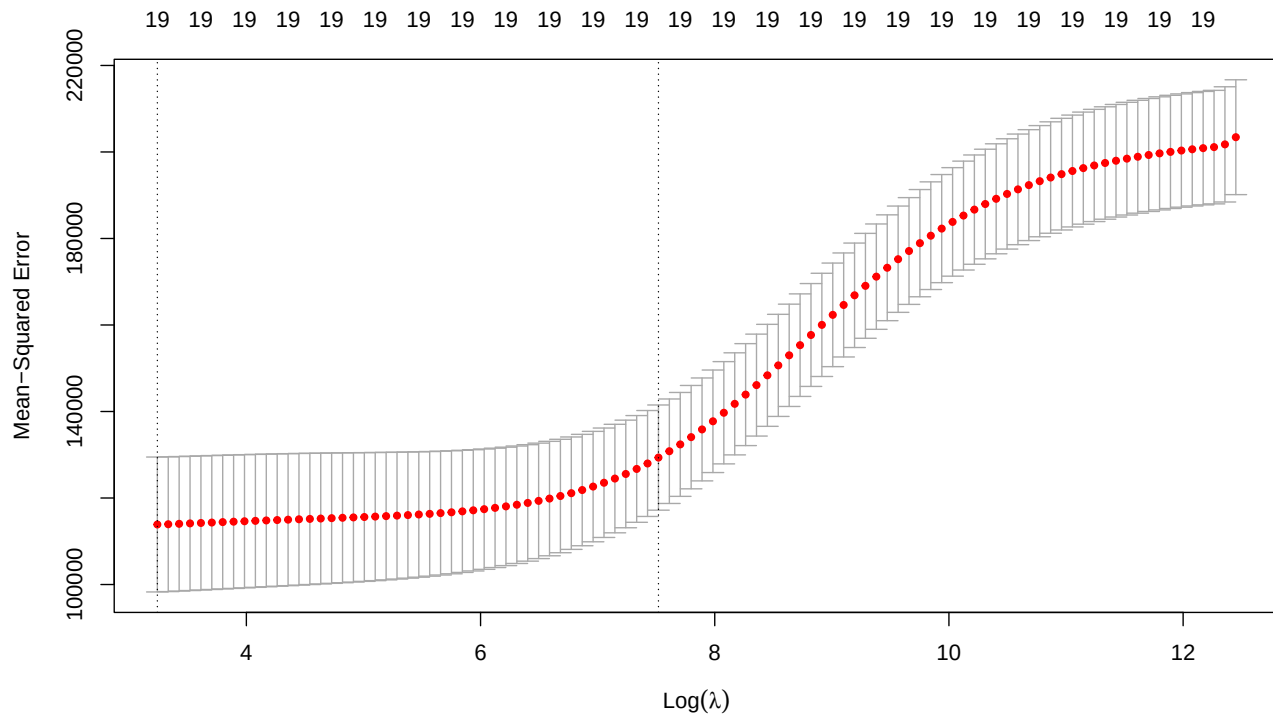
# First 10 Lambdas (remove the index to print all lambdas)
cbind("Lambda" = cv.10Fold$lambda,
      "10FCV MSE" = cv.10Fold$cvm)[1:10,]
```

	Lambda	10FCV MSE
[1,]	255282.1	203414.3
[2,]	232603.5	201753.4
[3,]	211939.7	201122.4
[4,]	193111.5	200881.5

```
[5,] 175956.0 200618.5
[6,] 160324.6 200331.2
[7,] 146081.8 200017.6
[8,] 133104.3 199675.6
[9,] 121279.7 199302.8
[10,] 110505.5 198896.5
```

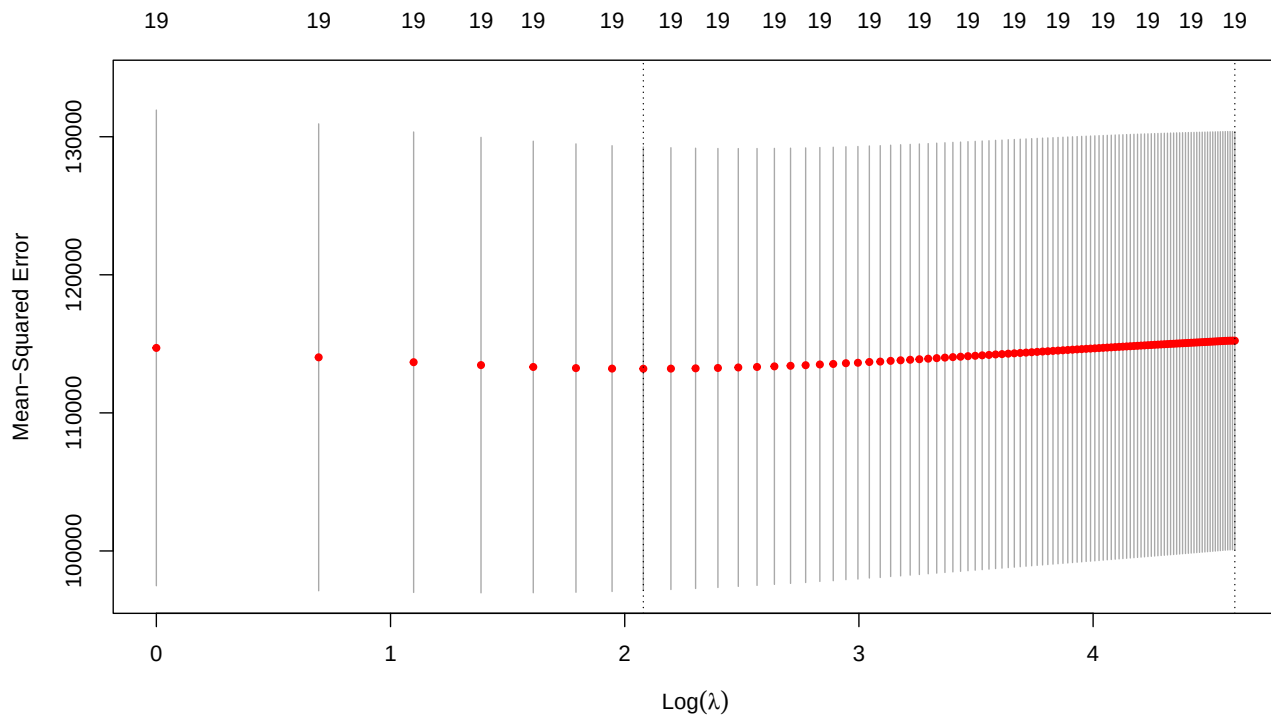
Let's plot all the Lambdas against their 10FCV MSE

```
plot(cv.10Fold)
```



The plot suggests that the MSE declines as the shrinkage lambda decreases. There is no clearly visible minimum in the graph. But we can evaluate this more carefully, by exploring several small lambdas, say from 1 to 100.

```
set.seed(1)
cv.10Fold <- cv.glmnet(x, y,
  alpha = 0,
  lambda = seq(0, 100)) # 10FCV is the default
plot(cv.10Fold)
```



The new plot shows some possible minimum value for the MSE around $\log(\lambda) = 2$. Let's find it more precisely, quantitatively. The best best Lambda is the one that minimizes the 10FCV test MSE. The `cv.glmnet()` output object has the attribute `lambda.min` that (conveniently) stores the Lambda with the smallest CV test MSE. Let's display it, along with its log to match the plot display. Let's also compute the corresponding CV test MSE by taking the CV vector 'cvm' (has all MSE's for all Lambdas) and finding its minimum value. I then display the results. And, sure enough, the best lambda is 8 with $\log(\lambda) = 2.079$ and 10FCV = 113,198.6:

```
best.lambda <- cv.10Fold$lambda.min
min.mse <- min(cv.10Fold$cvm)

cbind("Best Lambda" = best.lambda,
      "Log(Lambda)" = log(best.lambda),
      "Best 10FCV MSE" = min.mse)
```

```
Best Lambda Log(Lambda) Best 10FCV MSE
[1,]      8      2.079442      113198.6
```

Extracting Ridge Coefficients

You can extract Ridge regression coefficients for any level of Lambda shrinkage with the `coef()` function and the `s=` parameter.

Similar to OLS with `Lambda = 0`

```
options(scipen = 4)
ridge.0 <- round(coef(ridge.mod,
                      s = 0),
```

```
digits = 4)
```

For the optimal Lambda `s = best.lambda`.

```
ridge.best <- round(coef(ridge.mod,
                        s = best.lambda),
                    digits = 4)
```

For some arbitrary Lambda = 50

```
ridge.50 <- round(coef(ridge.mod,
                       s = 50),
                  digits = 4)
```

For some extreme shrinkage value, close to the Null model

```
ridge.huge <- round(coef(ridge.mod,
                        s = 10 ^ 20),
                    digits = 4)
```

Let's display all coefficients

```
ridge.all <- cbind(ridge.0,
                  ridge.best,
                  ridge.50,
                  ridge.huge)
colnames(ridge.all) <- c("OLS, Lambda = 0",
                        "Best Lambda = 8",
                        "Lambda = 50",
                        "Almost Null")

ridge.all
```

20 x 4 sparse Matrix of class "dgCMatrix"

	OLS, Lambda = 0	Best Lambda = 8	Lambda = 50	Almost Null
(Intercept)	81.1269	81.1269	48.2165	535.9259
AtBat	-0.6816	-0.6816	-0.3539	0.0000
Hits	2.7723	2.7723	1.9532	0.0000
HmRun	-1.3657	-1.3657	-1.2851	0.0000
Runs	1.0148	1.0148	1.1563	0.0000
RBI	0.7130	0.7130	0.8088	0.0000
Walks	3.3786	3.3786	2.7098	0.0000
Years	-9.0668	-9.0668	-6.2029	0.0000
CAtBat	-0.0012	-0.0012	0.0061	0.0000
CHits	0.1361	0.1361	0.1071	0.0000
CHmRun	0.6980	0.6980	0.6291	0.0000
CRuns	0.2959	0.2959	0.2173	0.0000
CRBI	0.2571	0.2571	0.2153	0.0000
CWalks	-0.2790	-0.2790	-0.1489	0.0000
LeagueN	53.2127	53.2127	45.8626	0.0000
DivisionW	-122.8345	-122.8345	-118.2304	0.0000
PutOuts	0.2639	0.2639	0.2502	0.0000
Assists	0.1699	0.1699	0.1208	0.0000
Errors	-3.6856	-3.6856	-3.2771	0.0000
NewLeagueN	-18.1051	-18.1051	-9.4235	0.0000

As you can see from the output above, the OLS coefficients are the same as those for the Ridge model with $\text{Lambda} = 8$ (minimal shrinkage). The coefficients appear to be identical, but that is due to rounding. With sufficient decimals you should see some very small differences. You can also see that an arbitrary value of $\text{Lambda} = 50$ biases the coefficients a little, but not by much. In contrast, the large Lambda yields an almost Null model. I say “almost” because Ridge does not drop the coefficients, and with sufficient decimals you would see very small non-zero coefficients. But for practical reasons, a $\text{Lambda} = 10^{20}$ yields very similar results to the Null model.

You can also get coefficients for a specific group of lambdas, any sequence of lambdas or a particular lambda number:

```
ridge.mod.group <- glmnet(x, y,
                          alpha = 0,
                          lambda = c(10 ^ 20, 10000, 10, 0))
round(coef(ridge.mod.group),
      digits = 4) # Take a look
```

```
20 x 4 sparse Matrix of class "dgCMatrix"
      s0      s1      s2      s3
(Intercept) 535.9259 392.7312 122.3737 161.9747
AtBat      0.0000  0.0411  -1.2068  -1.9484
Hits       0.0000  0.1545   4.2932   7.3497
HmRun      0.0000  0.5814  -0.4841   4.1744
Runs       0.0000  0.2574   0.4559  -2.2543
RBI        0.0000  0.2670   0.3972  -1.0056
Walks      0.0000  0.3236   4.4015   6.1740
Years      0.0000  1.2262 -10.7809  -2.8965
CAtBat     0.0000  0.0035  -0.0226  -0.1878
CHits      0.0000  0.0130   0.1652   0.2158
CHmRun     0.0000  0.0974   0.6898  -0.0523
CRuns      0.0000  0.0260   0.4727   1.4041
CRBI       0.0000  0.0269   0.3338   0.7620
CWalks     0.0000  0.0278  -0.4597  -0.7913
LeagueN    0.0000  0.1688  58.9833  63.5176
DivisionW  0.0000 -7.0644 -124.1991 -116.7595
PutOuts    0.0000  0.0186   0.2742   0.2814
Assists    0.0000  0.0029   0.2369   0.3767
Errors     0.0000 -0.0245  -3.8505  -3.4180
NewLeagueN 0.0000  0.3930 -25.4166 -25.6552
```

Or display coefficients for a sequence of Lambdas

```
ridge.mod.sequence <- glmnet(x, y,
                              alpha = 0,
                              lambda = seq(1, 5))
round(coef(ridge.mod.sequence), digits = 4) # Take a look
```

```
20 x 5 sparse Matrix of class "dgCMatrix"
      s0      s1      s2      s3      s4
(Intercept) 143.9106 150.3870 154.8083 159.1816 162.4634
AtBat      -1.5387  -1.6311  -1.7190  -1.8126  -1.9006
Hits       5.4047   5.6976   6.0451   6.4387   6.8769
HmRun      0.5463   0.7278   1.1689   1.7513   2.6002
Runs      -0.1905  -0.3778  -0.6670  -1.0199  -1.5030
RBI        0.0863   0.0408  -0.0849  -0.2531  -0.5051
Walks      5.0681   5.2665   5.4616   5.6794   5.9210
```

Years	-10.3670	-10.3179	-9.5292	-8.4591	-6.5952
CAtBat	-0.0495	-0.0580	-0.0739	-0.0942	-0.1270
CHits	0.1815	0.2031	0.2098	0.2159	0.2236
CHmRun	0.6451	0.7134	0.6671	0.5850	0.4177
CRuns	0.6622	0.7101	0.8042	0.9205	1.0939
CRBI	0.3905	0.3705	0.4008	0.4474	0.5344
CWalks	-0.5767	-0.6132	-0.6492	-0.6891	-0.7358
LeagueN	60.8330	61.4366	61.8941	62.3310	62.8899
DivisionW	-123.1024	-122.6512	-121.8053	-120.7689	-119.2272
PutOuts	0.2783	0.2796	0.2806	0.2815	0.2820
Assists	0.2799	0.2926	0.3072	0.3240	0.3458
Errors	-3.7777	-3.7555	-3.7034	-3.6377	-3.5495
NewLeagueN	-27.3382	-27.9432	-27.9504	-27.7946	-27.2580

And also display coefficients for a particular sequence order of Lambda

```
ridge.mod.sequence$lambda[3] # e.g., display the third Lambda
```

```
[1] 3
```

```
round(coef(ridge.mod.sequence)[,3],
      digits = 4) # Its coefficients
```

(Intercept)	AtBat	Hits	HmRun	Runs	RBI
154.8083	-1.7190	6.0451	1.1689	-0.6670	-0.0849
Walks	Years	CAtBat	CHits	CHmRun	CRuns
5.4616	-9.5292	-0.0739	0.2098	0.6671	0.8042
CRBI	CWalks	LeagueN	DivisionW	PutOuts	Assists
0.4008	-0.6492	61.8941	-121.8053	0.2806	0.3072
Errors	NewLeagueN				
-3.7034	-27.9504				

Predicting with Ridge

We can use the `predict()` function to obtain Ridge regression coefficients using the attribute `type = "coefficients"`. Let's find the Ridge regression coefficients for the best lambda:

```
predict(ridge.mod,
      s = best.lambda,
      type = "coefficients")
```

20 x 1 sparse Matrix of class "dgCMatrix"

```

      s1
(Intercept) 81.126931905
AtBat      -0.681595889
Hits       2.772311585
HmRun     -1.365680115
Runs      1.014826482
RBI       0.713022449
Walks     3.378557597
Years    -9.066800407
CAtBat   -0.001199478
```

CHits	0.136102881
CHmRun	0.697995816
CRuns	0.295889602
CRBI	0.257071062
CWalks	-0.278966595
LeagueN	53.212720343
DivisionW	-122.834451506
PutOuts	0.263887567
Assists	0.169879575
Errors	-3.685645337
NewLeagueN	-18.105095956

Now let's use the `predict()` function to make actual predictions. To do predictions with Ridge regression, you need to store the observations you want to do predictions for in a **new X** and specify it with the `newx =` parameter. This `newx` can either be new data for which you want to make predictions, or some test sub-sample to evaluate the model's accuracy. For the purposes of this example, let's just draw a random test sub-sample with 5% of the existing observations, using the best lambda we found above from the same `x` predictor matrix above.

```
set.seed(1)
test <- sample(1:nrow(x),
              0.05 * nrow(x))
x.test <- x[test, ] # Extract test sub-sample
ridge.pred <- predict(ridge.mod, # predict outcomes
                     s = best.lambda,
                     newx = x.test)
ridge.pred # Take a look
```

	s1
-Mike Heath	463.6768
-John Russell	385.6552
-Pete Incaviglia	307.2966
-Glenn Davis	776.5389
-Frank White	859.7506
-Rafael Santana	347.9036
-Chili Davis	680.8958
-Herm Winningham	213.7646
-Robby Thompson	390.6993
-Joe Carter	647.9247
-Spike Owen	344.9689
-Mike Easler	673.2900
-Chris Bando	320.4533

Ridge and Logistic Regression

You can fit a Ridge shrinkage model on a Logistic Regression simply by adding the parameter `family = "binomial"` to both functions, `glmnet()` and `cv.glmnet()`. Selecting the best lambda is identical to quantitative Ridge models. The coefficient interpretation in Ridge Logistic is the same as for Logistic Regression interpretation based on log-odds effects.

To illustrate Ridge Logistic, we can access the `SAHeart.data` dataset in Prof. Robert Tibshirani's web site, which contains variables associated with coronary heart disease. The dataset and variable descriptions are available at: <https://rdrr.io/cran/ElemStatLearn/man/SAheart.html>.

Once you have downloaded the data set, you can open it directly. I downloaded the file in CSV format with the name `Heart.csv`, which can be read as follows:

```
heart <- read.table("Heart.csv", sep = ",", head = T)
```

Once you have the data in the heart data frame object, the next step is to create the predictor matrix **x** and the outcome vector **y**. The outcome variable is binary, **chd** = **1** for disease, **0** otherwise)

```
x <- model.matrix(chd ~ ., data = heart)[-1]
y <- heart$chd
```

We then follow the same steps as for quantitative Ridge regressions, described above, with three differences:

- We use the **family = "binomial"** parameter
- The CV test reported in **\$cvm** is deviance using the $2LL = -2 * \text{Log-Likelihood}$ fit statistic for GLM models
- If we want to use **classification error** as the CV test deviance, we simply include the **measure.type = "class"** parameter.

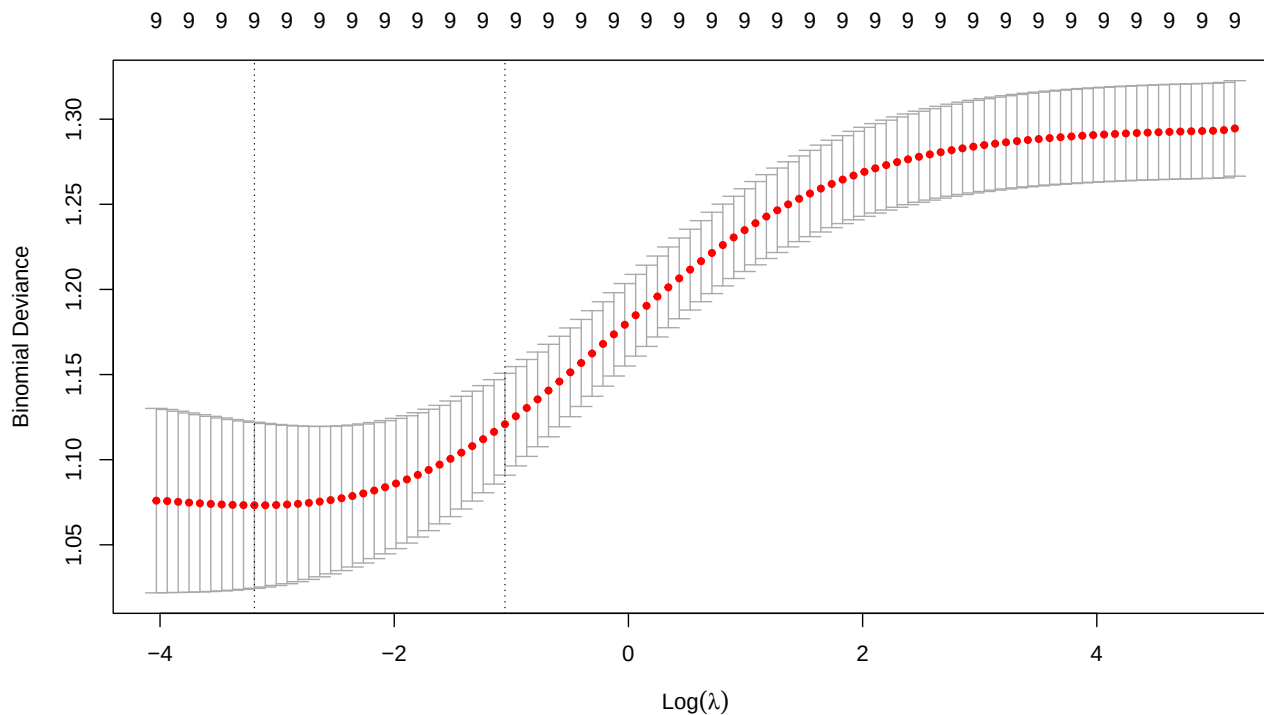
```
# Fit the Ridge Logistic model
ridge.logit=glmnet(x, y,
                  alpha = 0,
                  family = "binomial")

# Find best lambda:
set.seed(1)
cv.10Fold.logit <- cv.glmnet(x, y,
                           alpha = 0,
                           family = "binomial")

# Report (first 10 Lambdas to minimize the printout) and plot
cbind("Lambda" = cv.10Fold.logit$lambda,
      "10-Fold CV Deviance" = cv.10Fold.logit$cvm)[1:10,]
```

	Lambda	10-Fold CV Deviance
[1,]	177.45951	1.294554
[2,]	161.69449	1.293559
[3,]	147.33000	1.293179
[4,]	134.24161	1.293039
[5,]	122.31596	1.292886
[6,]	111.44974	1.292717
[7,]	101.54886	1.292533
[8,]	92.52753	1.292331
[9,]	84.30764	1.292109
[10,]	76.81798	1.291866

```
plot(cv.10Fold.logit)
```



Now let's find the **Best Lambda**.

```
best.lambda.logit <- cv.10Fold.logit$lambda.min
log(best.lambda) # Log to spot it in the plot
```

```
[1] 2.079442
```

```
min.mse.logit <- min(cv.10Fold.logit$cvm)
```

And then display the results

```
cbind("Best Lambda" = best.lambda.logit,
      "Log(Lambda)" = log(best.lambda.logit),
      "Best 10FCV Deviance" = min.mse.logit)
```

```
Best Lambda Log(Lambda) Best 10FCV Deviance
[1,] 0.04099545 -3.194294 1.073231
```

Then let's display coefficients

```
# Logit Model
ridge.logit.coef.0 <- coef(ridge.logit,
                           s = 0)
```

Alternatively, you can use the following to extract coefficients:

```
predict(ridge.logit, s = best.lambda.logit, type = "coefficients")
```

10 x 1 sparse Matrix of class "dgCMatrix"

```
              s1
(Intercept)  -5.318664789
sbp           0.0062893712
tobacco       0.0704757286
ldl           0.1421137485
adiposity     0.0167321735
famhistPresent 0.7664077959
typea        0.0280980909
obesity      -0.0376375020
alcohol       0.0003693287
age           0.0339089740
```

Best Lambda Model

```
ridge.logit.coef.best <- coef(ridge.logit,
                              s = best.lambda.logit)
```

Almost Null Model

```
ridge.logit.coef.large <- coef(ridge.logit,
                               s = 10 ^ 20)
```

Display all coefficients

```
all.coefs <- round(cbind(ridge.logit.coef.0,
                        ridge.logit.coef.best,
                        ridge.logit.coef.large),
                  digits = 4)
colnames(all.coefs) <- c("Logistic",
                        "Best Lambda",
                        "Almost Null Model")
all.coefs
```

10 x 3 sparse Matrix of class "dgCMatrix"

	Logistic	Best Lambda	Almost Null Model
(Intercept)	-5.7222	-5.3187	-0.6353
sbp	0.0065	0.0063	0.0000
tobacco	0.0754	0.0705	0.0000
ldl	0.1575	0.1421	0.0000
adiposity	0.0174	0.0167	0.0000
famhistPresent	0.8462	0.7664	0.0000
typea	0.0334	0.0281	0.0000
obesity	-0.0491	-0.0376	0.0000
alcohol	0.0002	0.0004	0.0000
age	0.0391	0.0339	0.0000

LASSO Regression

To fit LASSO regression models we follow the exact same steps as above, except that we use the parameter `alpha = 1` rather than 0.

We can also fit Elastic Net regression models using $0 < \alpha < 1$, which is a hybrid between Ridge and LASSO.

Again, the one difference between Ridge regression and LASSO is that coefficients can shrink significantly with Ridge regression, but they never become 0 (except when lambda is infinite). In contrast, some LASSO coefficients can shrink to 0. The more you shrink, the more coefficients drop out of the model. Let's re-create the x predictor matrix and y outcome vector for the Hitters data set.

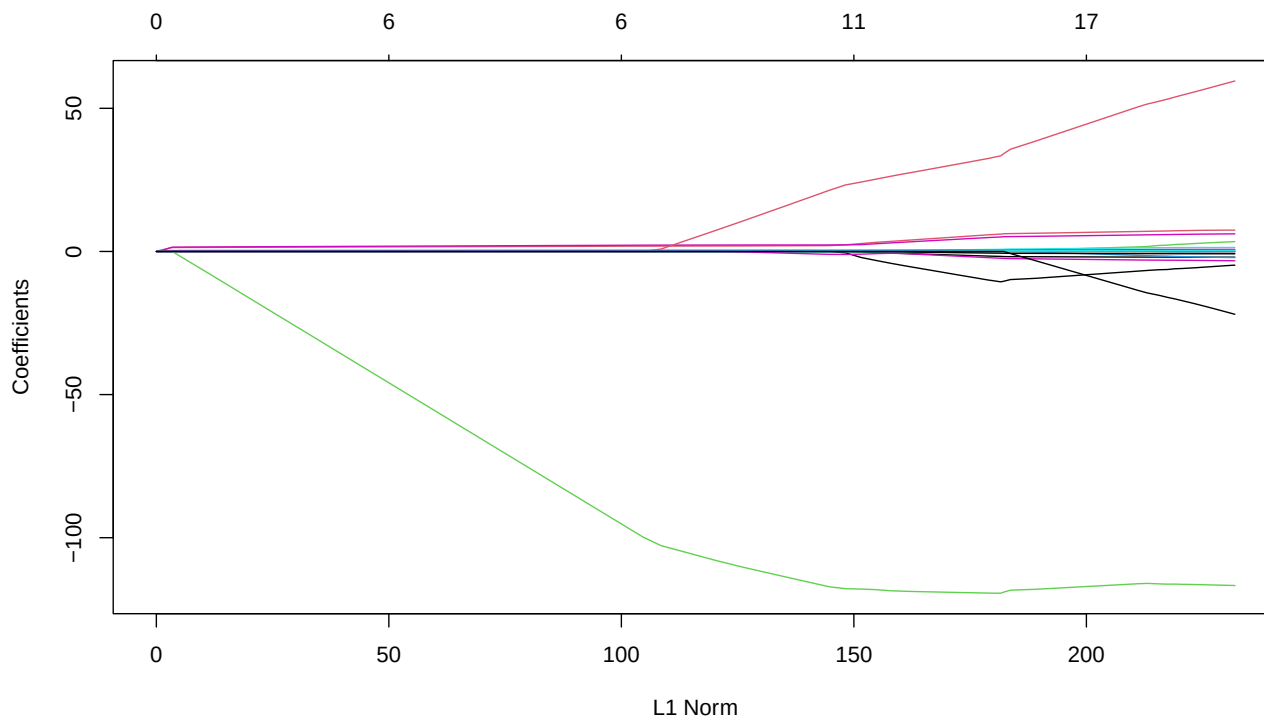
We already did this above, but let's repeat it here for convenience.

```
library(glmnet)
library(ISLR) # Contains the Hitters baseball player salary data set
Hitters <- na.omit(Hitters)
options(scipen = 4)

x <- model.matrix(Salary ~ ., data = Hitters)[ , -1]
y <- Hitters$Salary
```

Now let's fit the LASSO model and graph the **L1 Norm**. L1 Norm is the sum of the absolute values of the coefficients in a LASSO regression. Like L2 Norm, it measures the total amount of shrinkage obtained by a particular Lambda value used, but using absolute values, rather than squared values. The larger the L1 the smaller the shrinkage.

```
lasso.mod <- glmnet(x, y,
                    alpha = 1)
plot(lasso.mod) # Plot the L1 Norm against the coefficients
```



Optimal LASSO Lambda

```
set.seed(1) # To get repeatable results
cv.10Fold.lasso <- cv.glmnet(x, y,
                             alpha = 1) # 10-Fold is the default
# Use the parameter nfolds=xx to change the number of folds
```

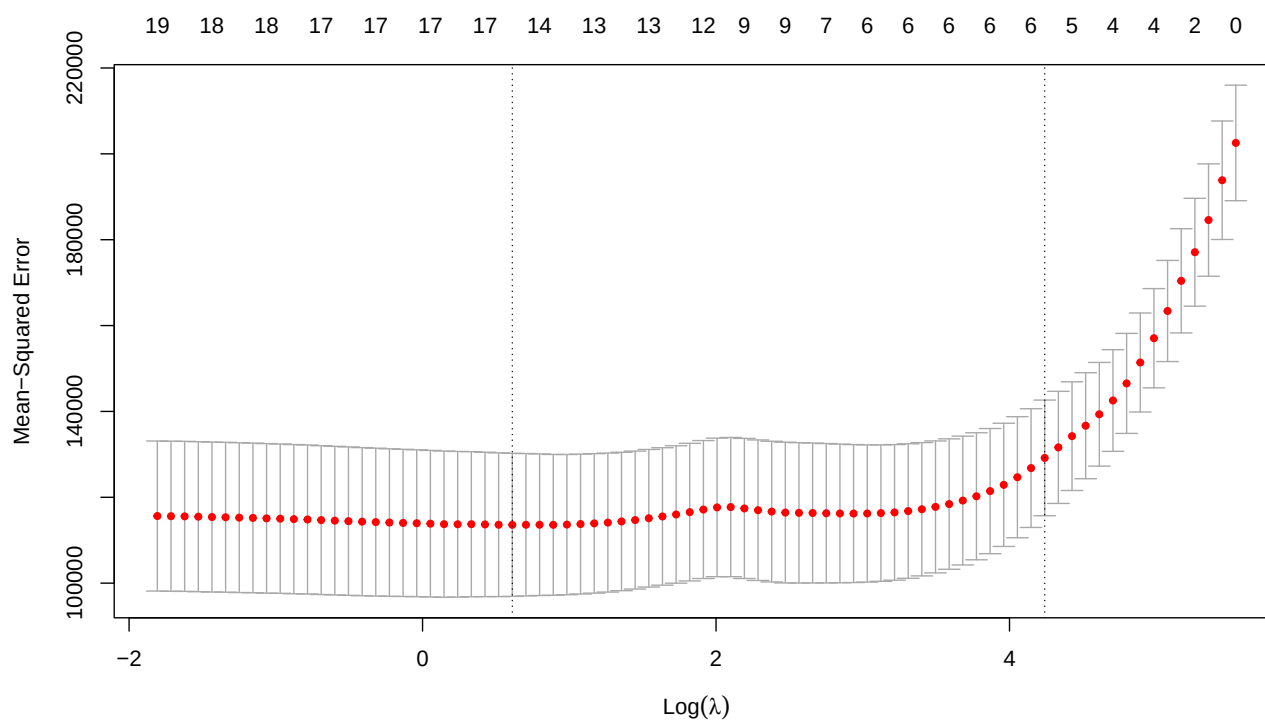
List all lambdas and corresponding CV test deviance (2LL)

```
round(cbind("Lambda" = cv.10Fold.lasso$lambda,
            "10 Fold 2LL" = cv.10Fold.lasso$cvm),
      digits = 4)[1:60, ]
```

	Lambda	10 Fold 2LL
[1,]	255.2821	202546.7
[2,]	232.6035	193851.4
[3,]	211.9397	184575.3
[4,]	193.1115	177082.0
[5,]	175.9560	170421.3
[6,]	160.3246	163395.7
[7,]	146.0818	157050.7
[8,]	133.1043	151398.1
[9,]	121.2797	146527.1
[10,]	110.5055	142542.5
[11,]	100.6885	139334.9
[12,]	91.7436	136669.2
[13,]	83.5934	134237.2
[14,]	76.1672	131622.7
[15,]	69.4007	129171.3
[16,]	63.2353	126797.2
[17,]	57.6177	124674.1
[18,]	52.4991	122897.6
[19,]	47.8352	121439.5
[20,]	43.5857	120233.3
[21,]	39.7136	119237.6
[22,]	36.1856	118422.7
[23,]	32.9710	117758.0
[24,]	30.0419	117215.0
[25,]	27.3731	116788.8
[26,]	24.9413	116481.7
[27,]	22.7256	116314.8
[28,]	20.7067	116229.1
[29,]	18.8672	116206.1
[30,]	17.1911	116232.5
[31,]	15.6639	116281.8
[32,]	14.2723	116336.6
[33,]	13.0044	116375.1
[34,]	11.8491	116442.3
[35,]	10.7965	116668.8
[36,]	9.8374	116994.1
[37,]	8.9634	117400.7
[38,]	8.1672	117720.2
[39,]	7.4416	117640.8
[40,]	6.7805	117145.3
[41,]	6.1782	116540.5

[42,]	5.6293	115999.3
[43,]	5.1292	115534.9
[44,]	4.6735	115119.2
[45,]	4.2584	114689.6
[46,]	3.8801	114367.9
[47,]	3.5354	114105.7
[48,]	3.2213	113915.2
[49,]	2.9351	113758.7
[50,]	2.6744	113640.1
[51,]	2.4368	113582.0
[52,]	2.2203	113610.7
[53,]	2.0231	113597.8
[54,]	1.8433	113581.6
[55,]	1.6796	113602.7
[56,]	1.5304	113674.8
[57,]	1.3944	113732.1
[58,]	1.2705	113713.6
[59,]	1.1577	113735.3
[60,]	1.0548	113850.6

```
plot(cv.10Fold.lasso) # Plot all lambdas vs. MSEs
```



Let's find the best lambda that minimizes 10FCV deviance

```
best.lambda.lasso <- cv.10Fold.lasso$lambda.min
log(best.lambda.lasso) # Spot it in the plot
```

```
[1] 0.6115809
```

```

min.mse.lasso <- min(cv.10Fold.lasso$cvm)

round(cbind("Best Lambda" = best.lambda.lasso,
           "Log(Lambda)" = log(best.lambda.lasso),
           "Best 10FCV MSE" = min.mse.lasso),
      digits = 4)

```

```

      Best Lambda Log(Lambda) Best 10FCV MSE
[1,]      1.8433      0.6116      113581.6

```

Extracting LASSO Coefficients

```

lasso.0 <- coef(lasso.mod, s = 0) # No shrinkage, same as OLS
lasso.best <- coef(lasso.mod, s = best.lambda.lasso) # best Lambda
lasso.50 <- coef(lasso.mod, s = 50) # Some lambda=50
lasso.large <- coef(lasso.mod, s = 10 ^ 20) # Null model

lasso.all <- cbind(lasso.0,
                  lasso.best,
                  lasso.50,
                  lasso.large)
colnames(lasso.all) <- c("OLS, Lambda = 0",
                        "Best Lambda = 1.84",
                        "Lambda = 50",
                        "Null")
lasso.all # Display all coefficients

```

```

20 x 4 sparse Matrix of class "dgCMatrix"
      OLS, Lambda = 0 Best Lambda = 1.84 Lambda = 50      Null
(Intercept) 162.4692458709      142.877615618      88.3914754 535.9259
AtBat      -2.0079033823      -1.793369131      .      .
Hits       7.4565103767       6.187727978      1.5867660      .
HmRun      3.4249665288       0.232913251      .      .
Runs      -2.1313920285      .      .      .
RBI       -0.7118808980      .      .      .
Walks      6.1480754543       5.147970767      1.8172405      .
Years     -4.8116447415     -10.392782086      .      .
CAtBat    -0.1377944362     -0.004469497      .      .
CHits     0.0801468485      .      .      .
CHmRun    -0.0005259073      0.585358718      .      .
CRuns     1.3906570825      0.763882254      0.1777376      .
CRBI      0.7192995638      0.388422191      0.3645924      .
CWalks    -0.8088092842     -0.629678348      .      .
LeagueN   59.5291502350     34.226933787      .      .
DivisionW -116.7266993092   -118.980715736   -43.2582890      .
PutOuts    0.2832108961      0.279042882      0.1344665      .
Assists    0.3529653782      0.223985943      .      .
Errors    -3.2537313982     -2.436300484      .      .
NewLeagueN -21.9457936788      .      .      .

```

Let's compare the effects for the best LASSO and Ridge models

```
best.both <- cbind(ridge.best,
                  lasso.best)
colnames(best.both) <- c("Ridge",
                        "LASSO")
best.both # Display all coefficients
```

20 x 2 sparse Matrix of class "dgCMatrix"

	Ridge	LASSO
(Intercept)	81.1269	142.877615618
AtBat	-0.6816	-1.793369131
Hits	2.7723	6.187727978
HmRun	-1.3657	0.232913251
Runs	1.0148	.
RBI	0.7130	.
Walks	3.3786	5.147970767
Years	-9.0668	-10.392782086
CAtBat	-0.0012	-0.004469497
CHits	0.1361	.
CHmRun	0.6980	0.585358718
CRuns	0.2959	0.763882254
CRBI	0.2571	0.388422191
CWalks	-0.2790	-0.629678348
LeagueN	53.2127	34.226933787
DivisionW	-122.8345	-118.980715736
PutOuts	0.2639	0.279042882
Assists	0.1699	0.223985943
Errors	-3.6856	-2.436300484
NewLeagueN	-18.1051	.

Notice again that the OLS and best LASSO models are just about the same. This is not surprising because the best Ridge model was also very close to the OLS model, suggesting that there are no major dimensionality issues, so naturally, the best LASSO model will be close to the OLS model.

Predicting with LASSO

As with Ridge regression, we can use the `predict()` function to obtain LASSO regression coefficients

```
lasso.coef <- predict(lasso.mod,
                     s = best.lambda.lasso,
                     type = "coefficients")
lasso.coef # Notice that the dropped predictors show blanks
```

20 x 1 sparse Matrix of class "dgCMatrix"

	s1
(Intercept)	142.877615618
AtBat	-1.793369131
Hits	6.187727978
HmRun	0.232913251
Runs	.
RBI	.
Walks	5.147970767
Years	-10.392782086
CAtBat	-0.004469497

CHits	.
CHmRun	0.585358718
CRuns	0.763882254
CRBI	0.388422191
CWalks	-0.629678348
LeagueN	34.226933787
DivisionW	-118.980715736
PutOuts	0.279042882
Assists	0.223985943
Errors	-2.436300484
NewLeagueN	.

To make actual predictions and to run LASSO Logistic models, the processes are identical to the ones for Ridge regression above.